

SQL Customer Behavior Analysis

An end-to-end relational database analysis investigating purchasing patterns, discount sensitivity, revenue segmentation, and demographic loyalty drivers in PostgreSQL.

 PostgreSQL 18

 pgAdmin 4

 3,900 Transactions

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Project Workflow & Architecture



1. Schema Inspection

Data profiling, structure verification, and checking column metrics.



2. Data Cleaning

Handling missing values, type casting, and standardizing categories.



3. Aggregations

Evaluating revenue drivers, promo impacts, and shipping methods.



4. CTE Segmentation

Categorizing users into New, Returning, and Loyal cohorts.



5. Business Strategy

Formulating actionable recommendations for executive teams.

Database Engine

Utilized PostgreSQL 18 with relational indexing for sub-millisecond aggregate processing.

Query Techniques

Aggregates, Subqueries, Window Functions, CASE Logic, and CTE constructs.

Core Objective

Extract actionable customer intelligence to optimize promotions and growth.



Dataset Schema & Inspection

Initial Data Exploration

Executing baseline data checks to review structural formatting, column data types, and initial sample records.

```
SELECT * FROM customer
LIMIT 20;
```

Key Attributes Profiled

Field Name	Data Type	Description
customer_id	INTEGER	Unique Identifier
purchase_amount	NUMERIC	Order Value (\$)
discount_applied	VARCHAR	Promo Indicator

pgAdmin 4

File Object Tools Edit View Window Help

Obj:

customer_behaviour/postgres@PostgreSQL 18

Properties X SQL X Statistics X Dependencies X Dependents X Processes X customer_behaviour/postgres@PostgreSQL 18*

Servers (1)
PostgreSQL 18
Databases (2)
customer_behaviour

Casts
Catalogs
Event Triggers
Extensions
Foreign Data W
Languages
Publications
Schemas (1)
public
Aggregate
Collations
Domains
FTS Confi
FTS Dictic
FTS Parse
FTS Temp
Foreign Te
Functions
Materializ
Operators

Query Query History

1 select * from customer limit 20
2
3 --1)What is the total revenue generated by male vs female customers?
4 select gender,sum(purchase_amount) as revenue
5 from customer
6 group by gender
7

Data Output Messages Notifications

Showing rows: 1 to 20 Page No: 1 of 1

	customer_id bigint	age bigint	gender text	item_purchased text	category text	purchase_amount bigint	location text	size text	color text	season text	review_rating double precision
1	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1
2	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1
3	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1
4	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5
5	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoi...	Spring	2.7
6	6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9
7	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2
8	8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2

Total rows: 20 Query complete 00:00:00.262 CRLF Ln 1, Col 32



Gender Revenue Distribution

\$157,890

MALE TOTAL REVENUE

\$75,191

FEMALE TOTAL REVENUE

Analytical Question

What is the total revenue generated by male vs female customer segments?

```
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer
GROUP BY gender;
```

Core Insight

Male customers generate **67.7%** of total platform revenue, representing a significant volume advantage that guides target marketing spend.

pgAdmin 4

File Object Tools Edit View Window Help

Obj Servers (1) PostgreSQL 18 Databases (2) customer_behav

Cast Catalogs Event Triggers Extensions Foreign Data W Languages Publications Schemas (1) public Aggregate Collations Domains FTS Confi FTS Dictic Aa FTS Parse FTS Temp Foreign Te Functions Materializ Operators

customer_behaviour/postgres@PostgreSQL 18

Query Query History

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1 select * from customer limit 20
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6 group by gender
7
```

Data Output Messages Notifications

	gender	revenue
1	Female	75191
2	Male	157890

Total rows: 2 Query complete 00:00:00.324 CRLF Ln 6, Col 16

High-Spender Discount Analysis

839 Users

HIGH-SPENDERS USING DISCOUNTS

Subquery Filtering Logic

Filtering transactions where customers applied discounts but still exceeded overall platform average transaction value.

```
SELECT * FROM customer
WHERE discount_applied = 'Yes'
AND purchase_amount > (
  SELECT AVG(purchase_amount) FROM customer
);
```

Business Finding

839 transactions used promotional discounts while spending above average (\$59.76), proving discounts successfully drive premium cart sizes without eroding revenue.

pgAdmin 4

File Object Tools Edit View Window Help

Obj

Servers (1)

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 - Materializ
 - Operators

customer_behaviour/postgres@PostgreSQL 18

group by gender

--2)Which customers used a discount but still spent more than the average purchase amount?

select customer_id,purchase_amount

from customer

where discount_applied = 'Yes' and purchase_amount > (select avg(purchase_amount) from customer)

Data Output

Showing rows: 1 to 839

Page No: 1 of 1

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90

Total rows: 839 | Query complete 00:00:00.260 | CRLF | Ln 11, Col 94



Top Product Rating Rankings

Top 5 Rated Items

Rank	Item Name	Avg Rating
#1	Gloves	★ 3.86
#2	Sandals	★ 3.84
#3	Boots	★ 3.82
#4	Hat	★ 3.80
#5	Skirt	★ 3.78

SQL Aggregation Method

```
SELECT item_purchased,  
ROUND(AVG(review_rating)::numeric, 2) AS avg_rating  
FROM customer  
GROUP BY item_purchased  
ORDER BY avg_rating DESC LIMIT 5;
```

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including Servers (1), PostgreSQL 18, Databases (2), and the selected database 'customer_behaviour'. The main pane shows a SQL query in the 'Query' tab:

```
--3)Which are the top 5 products with the highest average review rating?  
select item_purchased,round(avg(review_rating::numeric),2)as "average product rating"  
from customer  
group by item_purchased  
order by avg(review_rating) desc  
limit 5;
```

The 'Data Output' tab shows the results of the query:

item_purchased	average product rating	
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

The status bar at the bottom indicates 'Total rows: 5' and 'Query complete 00:00:00.277'.



Shipping Method Performance

\$60.48

EXPRESS SHIPPING AVG

\$58.46

STANDARD SHIPPING AVG

Shipping Comparison Query

Evaluating order values across fulfillment types to test if faster delivery correlates with higher customer spend.

```
SELECT shipping_type,
ROUND(AVG(purchase_amount)::numeric, 2) AS avg_spend
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

Key Finding

Express shipping yields a higher average order value (\$60.48 vs \$58.46), suggesting speed-sensitive buyers represent premium basket potential.

pgAdmin 4

File Object Tools Edit View Window Help

Obj...

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customer_behaviour/postgres@PostgreSQL 18

Query Query History

--4)Compare the average purchase amounts between standard and express shipping.
select shipping_type,
round(avg(purchase_amount),2)
from customer
where shipping_type in ('Standard','Express')
group by shipping_type

Data Output Messages Notifications

	shipping_type	round
	text	numeric
1	Standard	58.46
2	Express	60.48

Showing rows: 1 to 2 Page No: 1 of 1

Total rows: 2 Query complete 00:00:00.225 CRLF Ln 25, Col 23



Subscription Impact Assessment

\$170,432

NON-SUBSCRIBER TOTAL

\$62,649

SUBSCRIBER TOTAL

Analytical Summary

Subscriber Status	User Count	Avg Spend
Non-Subscribers	2,847	\$59.87
Subscribers	1,053	\$59.50

Strategic Gap

Average order size is identical (~\$59.70). Non-subscribers drive 73% of overall revenue due to volume, revealing an opportunity to convert casual shoppers.

pgAdmin 4

File Object Tools Edit View Window Help

Obj

Servers (1)

- PostgreSQL 18
 - Databases (2)
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- Cast
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customer_behaviour/postgres@PostgreSQL 18

Query Query History

--5)Do subscribed customers spend more? Compare avg spend and total revenue between
--between subscribers and non-subscribers.
select subscription_status,
count(customer_id) as total_customers,
round(avg(purchase_amount),2) as avg_spend,
round(sum(purchase_amount),2) as total_revenue
from customer
group by subscription_status
order by total_revenue,avg_spend desc;

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	subscription_status	total_customers	avg_spend	total_revenue
	text	bigint	numeric	numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Total rows: 2 Query complete 00:00:00.248 CRLF Ln 35, Col 39



Product Discount Rate Analysis

Top Promotion-Reliant Items

Products with the highest percentage of discounted purchases:

Hat (50%)

DISCOUNT RATE

Sneakers (49%)

DISCOUNT RATE

Conditional Aggregate SQL

```
SELECT item_purchased,  
ROUND(100.0 * SUM(CASE WHEN discount_applied='Yes'  
THEN 1 ELSE 0 END) / COUNT(*), 2)  
AS discount_rate_pct  
FROM customer  
GROUP BY item_purchased ORDER BY discount_rate_pct  
DESC;
```

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure: Servers (1) -> PostgreSQL 18 -> Databases (2) -> customer_behaviour. The main pane shows a SQL query in the 'Query' tab:

```
--6) Which 5 products have the highest percentage of purchases with discounts applied?  
select item_purchased,  
round(100*sum(case when discount_applied='Yes' then 1 else 0 end)/count(*),2) as discount_rate  
from customer  
group by item_purchased  
order by discount_rate desc  
limit 5;
```

The 'Data Output' tab shows the results of the query:

	item_purchased	discount_rate
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

The status bar at the bottom indicates 'Total rows: 5' and 'Query complete 00:00:00.200'.



Loyalty Tier Segmentation

3,116

LOYAL TIERS

701

RETURNING TIERS

83

NEW TIERS

CTE Segmentation Query

```
WITH customer_type AS (  
  SELECT customer_id, previous_purchases,  
  CASE  
    WHEN previous_purchases = 1 THEN 'New'  
    WHEN previous_purchases BETWEEN 2 AND 10 THEN  
      'Returning'  
    ELSE 'Loyal'  
  END AS customer_segment  
  FROM customer  
)  
SELECT customer_segment, COUNT(*) FROM customer_type  
GROUP BY customer_segment;
```

pgAdmin 4

File Object Tools Edit View Window Help

Obj

customer_behaviour/postgres@PostgreSQL 18*

Query Query History

```
--7)Segment customers into new,returning, and loyal based on their total  
--number of previous purchases,and show the count of each segment.  
47 with customer_type as (  
48   select customer_id,previous_purchases,  
49   case  
50     when previous_purchases=1 then 'New'  
51     when previous_purchases between 2 and 10 then 'Returning'  
52     else 'Loyal'  
53   end as customer_segment  
54   from customer  
55 )  
56 select customer_segment,count(*) as "number of customers"  
57   from customer_type  
58   group by customer_segment
```

Data Output Messages Notifications

	customer_segment text	number of customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Total rows: 3 Query complete 00:00:00.239 CRLF Ln 46, Col 3



Category & Regional Insights

Category Performance Analysis

Multi-column grouping evaluating item revenue distribution and order frequencies across retail categories.

The screenshot shows the pgAdmin 4 interface with a SQL query window. The query is designed to find the top 3 most purchased products within each category. It uses a CTE named 'item_counts' to calculate the total orders for each item, then ranks them by item_rank within each category. The final result is a table with columns: item_rank, category, item_purchased, and total_orders.

```
--8) What are the top 3 most purchased products within each category?  
WITH item_counts AS (  
    SELECT category,  
           item_purchased,  
           COUNT(customer_id) AS total_orders,  
           ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS item_rank  
    FROM customer  
    GROUP BY category, item_purchased  
)  
SELECT item_rank, category, item_purchased, total_orders  
FROM item_counts  
WHERE item_rank <= 3;
```

item_rank	category	item_purchased	total_orders	
1	Accessori...	Jewelry	171	
2	Accessori...	Sunglasses	161	
3	Accessori...	Belt	161	
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171

Total rows: 11 Query complete 00:00:00.356 CRLF Ln 72, Col 1

Repeated Purchase Analytics

Customers who are repeat buyers

The screenshot shows the pgAdmin 4 interface with a SQL query window. The query is designed to find customers who are repeat buyers (more than 5 previous purchases) and are also likely to subscribe. It uses a CTE named 'repeat_buyers' to calculate the count of previous purchases for each customer, then filters for customers with more than 5 previous purchases. The final result is a table with columns: subscription_status and repeat_buyers.

```
--Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?  
SELECT subscription_status,  
       COUNT(customer_id) AS repeat_buyers  
FROM customer  
WHERE previous_purchases > 5  
GROUP BY subscription_status;
```

subscription_status	repeat_buyers
No	2518
Yes	958

Total rows: 2 Query complete 00:00:00.328 CRLF Ln 79, Col 30

Revenue contribution analysis by each category

62143

Young adult

59197

Middle-aged

55978

Adult

83

NSenior

pgAdmin 4

File Object Tools Edit View Window Help

Object Properties SQL Statistics Dependencies Dependents Processes customer_behaviour/postgres@PostgreSQL 18*

customer_behaviour/postgres@PostgreSQL 18

No limit

Query Query History

```
81
82 --Q10. What is the revenue contribution of each age group?
83 SELECT
84     age_group,
85     SUM(purchase_amount) AS total_revenue
86 FROM customer
87 GROUP BY age_group
88 ORDER BY total_revenue desc;
89
```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Total rows: 4 Query complete 00:00:00.270 CRLF Ln 88, Col 29



Strategic Recommendations

1. Male Demographic Targeting

Since male customers contribute \$157.8K (67.7% of total spend), focus digital acquisition budgets towards core male product lines.

2. Revamp Subscription Value

Subscriber average order value (\$59.50) is identical to non-subscribers. Introduce exclusive perks or tier discounts to boost membership spend.

3. Discount Optimization

839 high spenders still make premium purchases with promo codes. Maintain threshold discounts (\$50+ minimum spend) to prevent margin dilution.

4. Loyalty Retention Programs

3,116 users are classified as Loyal (10+ past orders). Implement a VIP reward tier to safeguard high-LTV customer retention.